FUTURE INTERNS

Data Science & Analytics Internship | Task 2 | FUTURE_DS_02

Customer Retention & Churn Analysis

A comprehensive analysis of churn patterns, retention drivers, and actionable recommendations for a subscription business

1,000	14.3%	32.1 mo	Rs 594
Total Customers	Churn Rate	Avg Tenure (Active)	Avg Monthly Charge

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Tools: Python | Pandas | Matplotlib | Seaborn | Plotly

1. Executive Summary

This report presents a comprehensive churn analysis of 1,000 customers from a subscription-based business. The overall churn rate stands at 14.3%, meaning 143 customers have discontinued their subscriptions. The analysis reveals that churn is heavily concentrated in the Basic plan, East region, and among customers within their first 24 months of tenure.

Key findings indicate that price-related concerns drive nearly half of all customer exits, while long-tenure customers demonstrate strong loyalty. Targeted interventions in early lifecycle engagement, plan value communication, and regional support quality can significantly reduce churn and improve customer lifetime value.

2. Dataset Overview

Metric	Value	Metric	Value
Total Customers	1,000	Churned Customers	143 (14.3%)
Retained Customers	857 (85.7%)	Avg Tenure — Active	32.1 months
Avg Tenure — Churned	19.4 months	Avg Monthly Charge — Active	Rs 594
Avg Monthly Charge — Churned	Rs 472	Tenure Range	1 to 60 months

3. Churn Analysis

3.1 Churn by Plan Type

Plan type is the strongest predictor of churn. Basic plan subscribers exit at nearly three times the rate of Premium customers, indicating a significant perceived value gap at the entry tier.

Plan	Total Customers	Churned	Churn Rate
Basic	401	84	20.9%
Standard	351	41	11.7%
Premium	248	18	7.3%

3.2 Churn by Tenure Group

Customers in their first 24 months face the highest attrition risk. After month 25, churn drops dramatically, suggesting that customers who survive the initial period develop strong loyalty.

Tenure Group	Total Customers	Churned	Churn Rate
0-12 Months	249	58	23.3%
13-24 Months	245	51	20.8%
25-36 Months	243	19	7.8%
37-60 Months	263	27	10.3%

3.3 Churn by Region

The East region leads in churn at 18.3%, nearly 70% higher than the South region (10.8%). This geographic disparity warrants dedicated regional investigations into service quality and competitive pressure.

Region	Total Customers	Churned	Churn Rate
East	257	47	18.3%
North	237	36	15.2%
West	265	34	12.8%
South	241	26	10.8%

3.4 Churn by Payment Method

UPI and Debit Card users show the highest churn rates, while Credit Card users are the most stable. This may reflect income stability and commitment levels associated with different payment preferences.

Payment Method	Total Customers	Churned	Churn Rate
UPI	243	38	15.6%
Debit Card	258	40	15.5%
Net Banking	251	36	14.3%
Credit Card	248	29	11.7%

4. Churn Reasons Analysis

Of the 143 customers who churned, all provided a reason for their exit. Price-related factors dominate, accounting for over 45% of total churn:

Churn Reason	Customers	% of Churned
Better Price Elsewhere	37	25.9%
Financial Constraints	28	19.6%
Poor Customer Service	25	17.5%
Lack of Features	22	15.4%
Moved Away	16	11.2%
Switched to Competitor	15	10.5%

Key observation: "Better Price Elsewhere" is the single largest churn driver (25.9%), followed by "Financial Constraints" (19.6%). Together, price-related reasons account for 45.5% of all exits — pointing to a competitive pricing gap rather than a product quality issue.

5. Key Insights

1 Basic Plan has 3x higher churn than Premium (20.9% vs 7.3%)

The Basic tier is severely underserving its subscribers. Customers on this plan feel they are not receiving sufficient value relative to their spend. This gap needs to be closed through feature additions or loyalty incentives.

2 First 24 months are the highest-risk window (20-23% churn)

Churn drops to 7-10% after month 25, suggesting that customers who survive the initial period develop strong loyalty. A structured early-lifecycle intervention program could save 40-50 customers annually.

3 Price sensitivity drives 45% of all churn

"Better Price Elsewhere" and "Financial Constraints" together account for nearly half of all exits. This is a competitive market signal — the product is not differentiated enough to justify its price for price-sensitive segments.

4 East region underperforms by 70% vs South (18.3% vs 10.8%)

The East region shows consistently higher churn across all plan types. Possible causes include stronger local competition, weaker customer support, or demographic differences in price sensitivity.

5 Churned customers paid significantly less (Rs 472 vs Rs 594/mo)

This rules out bill shock as a churn cause. Lower-paying customers are on cheaper plans with less perceived value — they are underserved, not overcharged.

6 October 2023 cohort has the highest churn at 21.9%

A spike in the October cohort may indicate a product issue, poor onboarding experience, or an aggressive acquisition campaign that attracted low-intent customers. Retroactive review of that cohort is recommended.

6. Business Recommendations

01 **Early Lifecycle Intervention**
Deploy automated check-in emails and in-app nudges for customers in months 1-24. Flag customers with 0-2 products and high support calls for proactive outreach by a retention specialist.

02 **Basic Plan Value Upgrade**
Introduce a "Basic Plus" tier or add loyalty benefits (extra features, priority support) for Basic plan users after 12 months to reduce the 20.9% churn rate.

03 **Competitive Price Response Program**
Create a dedicated win-back campaign for customers citing "Better Price Elsewhere." Offer a 3-month discounted rate or price-match guarantee to retain price-sensitive users.

04 **East Region CX Audit**
Conduct a full customer experience audit in the East region. Investigate whether the 18.3% churn rate is driven by product gaps, support quality issues, or competitive market dynamics.

05 **Flexible Payment & Financial Hardship Program**
Offer EMI options, hardship deferrals, and payment holiday programs to reduce churn from "Financial Constraints." Target UPI and Debit Card users who show higher churn rates.

06 **Feature Gap Roadmap via Quarterly Surveys**
Launch a quarterly product survey for Basic and Standard plan users to build a prioritized feature roadmap addressing the 15.4% who churned due to "Lack of Features."

7. Conclusion

This analysis reveals that customer churn in this subscription business is not random — it is concentrated, predictable, and actionable. The Basic plan, early-tenure customers, and the East region together account for a disproportionate share of exits.

The dominant churn driver is price sensitivity, not product dissatisfaction, which means retention improvements can be achieved through pricing strategy, plan value communication, and targeted financial support programs. By implementing the six recommendations in this report, the business can realistically reduce overall churn by 30-40%, improve customer lifetime value, and strengthen its competitive positioning in high-churn regions.